Oracle goal-image rotation with frozen LeWM CEM on Push-T

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Abstract

The frozen LeWorldModel planner that reaches a dataset goal image on Push-T does not rotate the T in place when that same cost is given a rendered frame at $\theta + 90°$ or $\theta + 180°$. On 20 held-out starts the reach rate is 0.00 at a 50-step budget for both turns, and 0.00 for 180° at a 100-step budget. The rendered target is a real rotation: its latent is far from the start, and the block barely moves between the two renders. A network that predicts the goal latent was not trained.

1 Introduction

The official image-goal protocol, with actions z-scored from the Push-T training column, succeeds on 45 of 50 episodes [Maes et al., 2026]. A probe that reads angle from z is accurate near 180°, but a planning cost written on that probe does not turn the block. This note replaces the probe with an oracle: the goal latent is the encoding of a frame rendered at the desired angle and the same block position.

2 Methods

The encoder is the frozen checkpoint `quentinll/lewm-pusht`. The planner is the one in `scripts/eval_official_cem.py`. Actions are normalized with mean $[-0.00781, 0.00686]$ and unbiased standard deviation $[0.20847, 0.20675]$, then inverse-transformed and clipped to $[-1, 1]$ before `env.step`. CEM uses 300 samples, 30 iterations, and 30 elites. The horizon is 5 and the action block is 5. Pixel history length is 1. Each of 20 test episodes from `pusht_theta_probe.npz`, split with seed 0, is reset to its recorded agent and block pose. The goal frame is produced by `_set_state` with the same block position, angle increased by $\pi/2$ or π, and agent velocity 0. The start state is restored, and CEM minimizes $\|z_H - z_g\|^2$. Budgets are 50 environment steps for both turns and 100 steps for 180° only. A trial reaches the oracle when the wrapped angle error is below 20 degrees and the block has moved less than 15 pixels. Push-T does not expose a coverage score in the info dict.

Table 1: In-place oracle CEM on 20 held-out starts.

Turn	budget	mean $\lvert\Delta\theta\rvert$	mean Δxy	reach	wall (s)
+90°	50	91.94	27.76	0.00	15.56
+180°	50	153.05	50.46	0.00	15.48
+180°	100	151.60	53.39	0.00	30.97

3 Results

The goal render matches the command. Mean residual of the rendered angle is 0.023 degrees for +90° and 0.065 degrees for +180°. Mean block motion during that render is 0.19 and 0.12 pixels. Mean $\|z_g - z_t\|$ is 15.69 for +90° and 18.33 for +180°. Table 1 is the CEM outcome. No cell has a nonzero reach rate. Lengthening the 180° budget from 50 to 100 steps changes the mean angle error from 153.05 degrees to 151.60 degrees.

4 Discussion

The 25-step dataset goal protocol was 90 percent successful with this planner. In-place rotation with a true goal image is not. CEM cannot execute in-place 180° even with a true goal image under this horizon and budget. A map from $(z_t, \text{command})$ to z_g was therefore not fit. The next change is a longer horizon or a multi-stage plan, not a better text encoder.

5 Conclusion

Oracle goal images of a rotated T are easy to render and easy to tell apart in latent space. The working image-goal CEM does not drive the live block onto those images within 50 or 100 steps while staying in place. The language port should not be asked to invent a goal latent until planning can follow one that is given.

References

Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. Leworldmodel: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv preprint arXiv:2603.19312*, 2026. URL https://arxiv.org/abs/2603.19312.